



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 21.07.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापलंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 20.07.2024.

महोदय/Dear Sir,

आईईजीसी-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 20-जुलाई-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 20.07.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day Date of Reporting: 21-Jul-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	80633	58739	42824	27827	3552	213575
Peak Shortage (MW)	1714	478	0	419	0	2611
Energy Met (MU)	1890	1383	972	638	70	4952
Hydro Gen (MU)	391	49	119	78	36	672
Wind Gen (MU)	6	99	304	-	-	408
Solar Gen (MU)*	138.19	68.27	71.44	2.05	0.76	281
Energy Shortage (MU)	9.83	1.80	0.00	3.82	0.00	15.45
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	83771	59941	44358	29518	3577	214749
Time Of Maximum Demand Met	14:15	19:25	08:28	23:06	19:57	19:45

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.044	0.05	1.89	8.68	10.61	79.54	9.85

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	15918	0	363.2	215.1	1.7	340	0.00
	Haryana	13571	0	287.7	210.4	0.9	266	1.46
	Rajasthan	15459	0	331.6	127.1	4.2	791	3.92
	Delhi	7194	0	144.7	120.1	0.6	160	0.00
	UP	29041	386	601.7	293.5	0.7	432	3.62
	Uttarakhand	2449	0	53.8	26.9	0.5	222	0.75
	HP	1887	0	40.5	3.0	0.2	139	0.00
	J&K(UT) & Ladakh(UT)	2602	36	54.8	31.8	-2.9	111	0.08
	Chandigarh	395	0	7.7	7.7	0.0	0	0.00
WR	Railways_NR ISTS	202	0	4.3	3.5	0.8	98	0.00
	Chhattisgarh	5324	0	122.7	77.5	-2.9	224	0.00
	Gujarat	18842	0	417.0	152.5	-2.3	569	0.00
	MP	12564	0	274.9	150.7	0.1	689	1.80
	Maharashtra	22334	0	494.1	155.7	-8.6	884	0.00
	Goa	635	0	12.8	12.5	0.3	85	0.00
	DNHDDPDCL	1289	0	29.8	29.9	-0.1	50	0.00
	AMNSIL	836	0	18.8	6.1	0.2	289	0.00
	BALCO	529	0	12.6	12.3	0.3	144	0.00
SR	Andhra Pradesh	8201	0	182.0	31.7	-1.4	315	0.00
	Telangana	9801	0	192.7	113.3	-0.7	614	0.00
	Karnataka	9702	0	181.6	-37.2	-1.7	751	0.00
	Kerala	3903	0	76.6	43.4	0.1	347	0.00
	Tamil Nadu	16126	0	329.3	108.1	-1.9	1214	0.00
	Puducherry	426	0	9.7	9.2	-0.1	42	0.00
	Bihar	7702	119	163.0	152.3	-1.1	452	2.60
ER	DVC	3422	0	74.3	-35.3	-0.9	246	0.00
	Jharkhand	2141	0	46.1	32.9	0.5	365	1.22
	Odisha	5557	0	116.6	35.0	-1.8	386	0.00
	West Bengal	11027	0	236.3	95.4	-2.4	257	0.00
	Sikkim	79	0	1.3	1.3	-0.1	27	0.00
	Railways_ER ISTS	17	0	0.2	0.2	0.0	3	0.00
NER	Arunachal Pradesh	175	0	2.6	2.6	-0.5	30	0.00
	Assam	2370	0	46.6	39.4	0.2	149	0.00
	Manipur	179	0	3.1	3.0	0.1	94	0.00
	Meghalaya	311	0	5.6	-1.1	0.0	91	0.00
	Mizoram	125	0	2.5	0.7	0.3	41	0.00
	Nagaland	174	0	3.2	2.8	0.0	19	0.00
	Tripura	320	0	6.4	5.6	0.5	66	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	36.6	17.0	-24.8	-28.6
Day Peak (MW)	1963.7	783.1	-1024.0	-1372.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	447.2	-260.0	-134.3	-57.6	4.8	0.0
Actual(MU)	465.1	-236.2	-170.9	-64.6	2.2	-4.4
O/D/U/D(MU)	18.0	23.8	-36.6	-7.0	-2.6	-4.4

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	3827	12904	6438	5876	280	29324	56
State Sector	3896	8436	8983	1324	157	22795	44
Total	7723	21339	15421	7200	437	52119	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	891	1439	540	683	15	3568	67
Lignite	32	11	50	0	0	93	2
Hydro	391	49	119	78	36	672	13
Nuclear	30	62	76	0	0	168	3
Gas, Naptha & Diesel	36	47	5	0	21	109	2
RES (Wind, Solar, Biomass & Others)	151	168	398	4	1	722	14
Total	1531	1776	1188	765	73	5333	100
Share of RES in total generation (%)	9.88	9.46	33.48	0.51	1.05	13.53	
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	37.37	15.71	49.88	10.72	50.08	29.29	

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.029
Based on State Max Demands	1.084

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	213649	14:53	0
Non-Solar hr	214749	19:45	1243

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 07:00 to 17:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 21-Jul-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	701	0.0	17.0	-17.0
2	HVDC	PUSAULI B/B	-	2	97	0.0	2.2	-2.2
3	765 kV	GAYA-VARANASI	2	780	559	0.0	5.2	-5.2
4	765 kV	SASARAM-FATEHPUR	1	259	300	0.0	2.6	-2.6
5	765 kV	GAYA-BALIA	1	0	1002	0.0	15.5	-15.5
6	400 kV	PUSAULI-VARANASI	1	27	85	0.0	0.8	-0.8
7	400 kV	PUSAULI -ALLAHABAD	1	0	91	0.0	1.4	-1.4
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	773	0.0	11.3	-11.3
9	400 kV	PATNA-BALIA	2	0	531	0.0	7.1	-7.1
10	400 kV	NAUBATPUR-BALIA	2	121	137	0.0	1.5	-1.5
11	400 kV	BIHARSHARIFF-BALIA	2	277	139	0.4	0.0	0.4
12	400 kV	MOTIHARI-GORAKHPUR	2	0	403	0.0	5.5	-5.5
13	400 kV	BIHARSHARIFF-VARANASI	2	69	268	0.0	4.1	-4.1
14	220 kV	SAHUPURI-KARAMNASA	1	37	58	0.0	0.5	-0.5
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.7	0.0	0.7
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						1.0	74.6	-73.5
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	0	1668	0.0	26.8	-26.8
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	2111	0	35.1	0.0	35.1
3	765 kV	JHARSUGUDA-DURG	2	200	425	0.0	0.6	-0.6
4	400 kV	JHARSUGUDA-RAIGARH	4	0	431	0.0	7.2	-7.2
5	400 kV	RANCHI-SIPAT	2	550	0	7.5	0.0	7.5
6	220 kV	BUDHIPADAR-RAIGARH	1	7	67	0.0	0.7	-0.7
7	220 kV	BUDHIPADAR-KORBA	2	0	105	0.0	1.6	-1.6
ER-WR						42.6	36.9	5.7
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	757	0	13.4	0.0	13.4
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	393	0.0	8.5	-8.5
3	765 kV	ANGUL-SRIKAKULAM	2	0	1647	0.0	21.4	-21.4
4	400 kV	TALCHER-I/C	2	1066	0	23.2	0.0	23.2
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						13.4	29.9	-16.6
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	19	381	0.0	4.6	-4.6
2	400 kV	ALIPURDUAR-BONGAIGAON	2	59	540	0.0	5.9	-5.9
3	220 kV	ALIPURDUAR-SALAKATI	2	0	108	0.0	1.5	-1.5
ER-NER						0.0	12.1	-12.1
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	0	504	0.0	11.1	-11.1
NER-NR						0.0	11.1	-11.1
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	4526	0.0	86.2	-86.2
2	HVDC	VINDHYACHAL B/B	-	0	106	2.4	0.0	2.4
3	HVDC	MUNDRA-MOHINDERGARH	2	0	2356	0.0	60.4	-60.4
4	765 kV	GWALIOR-AGRA	2	0	2999	0.0	53.1	-53.1
5	765 kV	GWALIOR-PHAGI	2	0	1911	0.0	31.5	-31.5
6	765 kV	JABALPUR-ORAI	2	0	1521	0.0	57.3	-57.3
7	765 kV	GWALIOR-ORAI	1	784	0	14.6	0.0	14.6
8	765 kV	SATNA-ORAI	1	0	1143	0.0	23.1	-23.1
9	765 kV	BANASKANTHA-CHITORGARH	2	770	852	2.0	9.3	-7.2
10	765 kV	VINDHYACHAL-VARANASI	2	0	3929	0.0	76.2	-76.2
11	400 kV	ZERDA-KANKROLI	1	186	181	0.6	1.8	-1.2
12	400 kV	ZERDA -BHINMAL	1	128	125	0.4	1.3	-0.9
13	400 kV	VINDHYACHAL -RIHAND	1	968	0	21.5	0.0	21.5
14	400 kV	RAPP-SHUJALPUR	2	122	692	0.1	9.7	-9.5
15	400 kV	NEEMUCH-Chittorgarh	2	147	737	0.1	9.9	-9.8
16	220 kV	BHANPURA-RANPUR	1	0	113	0.0	2.1	-2.1
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.0	-2.0
18	220 kV	MEHGAON-AURAIYA	1	65	20	0.4	0.1	0.3
19	220 kV	MALANPUR-AURAIYA	1	38	37	0.1	0.3	-0.2
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						42.3	424.1	-381.8
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	998	0	23.4	0.0	23.4
2	HVDC	RAIGARH-PUGALUR	2	865	0	32.2	0.0	32.2
3	765 kV	SOLAPUR-RAICHUR	2	2859	0	44.4	0.0	44.4
4	765 kV	WARDHA-NIZAMABAD	2	882	1436	4.9	5.5	-0.6
5	765 kV	WARORA-WARANGAL(NEW)	2	1016	1269	5.9	4.5	1.5
6	400 kV	KOLHAPUR-KUDGI	2	2480	0	48.0	0.0	48.0
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	29	1.3	0.0	1.3
9	220 kV	XELDEM-AMBEWADI	1	0	103	1.8	0.0	1.8
WR-SR						162.0	10.0	152.0

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)		
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	772	683	699	16.78	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	916	0	840	20.16	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	148	86	-101	-2.43	
	NER	132kV GELEPHU-SALAKATI	40	20	35	0.83	
	NER	132kV MOTANGA-RANGIA	69	39	53	1.26	
NEPAL	NR	NEPAL IMPORT (FROM UP)	-53	0	0	0.00	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	65	0	44	1.05	
	ER	NEPAL IMPORT (FROM BIHAR)	0	0	0	0.00	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	718	613	665	15.97	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-935	-934	-935	-22.99	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1372	-1006	-1193	-28.64	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-89	0	-75	-1.80	

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nepal	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.15
Bangladesh	22.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.28
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	22.43	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.43

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL	
		BILATERAL TOTAL	COLLECTIVE							
			IDAM			RTM				
			IEX	PXIL	HPX	IEX	PXIL	HPX		
Bhutan	37.57	0.00	1.46	0.00	0.00	0.00	0.00	0.00	39.03	
Nepal	2.63	0.00	13.18	0.00	0.00	0.00	0.28	0.00	0.00	16.09
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	40.20	0.00	14.64	0.00	0.00	0.00	0.28	0.00	0.00	55.12

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	37.57	0.00	1.46	0.00	0.00	0.00	0.00	0.00	39.03
Nepal	2.48	0.00	13.18	0.00	0.00	0.28	0.00	0.00	15.94
Bangladesh	-22.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-22.28
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	17.77	0.00	14.64	0.00	0.00	0.28	0.00	0.00	32.69